

DP Tradeking HRMS & Payroll System

By

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Under Guidance of

Internal Guide

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AND
TECHNOLOGY
Changa

Acknowledgement

Knowledge in itself is a continuous process. At this moment of our substantial enhancement, We rarely find words to express our gratitude towards those who were constantly involved with us.

The completion of any inter disciplinary project depends upon coordination, cooperation and combined efforts of several resources of knowledge, creativity, skill, energy and time. The work being accomplished now, we feel our sincerest urge to recall and knowledge through these lines, trying our best to give full credit wherever it deserves.

We would like to thank our project guide **Dr. Mittal Desai**, Principal **Dr.Dharmendra Patel** and Dean **Dr. Sanskruti Patel** who advised and gave us moral support through the duration of our project. Without their constant encouragement we could not have been able to achieve what we have.

It's our good fortune that we had support and well wishes of many. We are thankful to all and those names which have been forgotten to acknowledge here but contributions have not gone unnoticed.

With Sincere Regards,

Sujal Patel(24MCA074)

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1 Company and Project Profile

Category	Description
Company Name	DP Tradeking
Industry Type	Trading and Business Operations Management
Company Objective	To improve operational efficiency through structured workforce management and digital transformation.
Current Challenges	Manual employee records, attendance errors, lack of centralized database, limited data security, and no role-based access control.
System Requirement	A secure, web-based HR Management System to automate employee, department, and attendance management.
Project Title	DP Tradeking HRMS & Payroll
Project Type	Web-Based Enterprise Application
Technology Stack	MongoDB (Database), Express.js (Backend), React.js (Frontend), Node.js (Runtime Environment)
Authentication Mechanism	JWT-based role authentication (Admin, HR, Employee)
Main Functional Modules	Authentication Module, Employee Management Module, Department Module, Attendance Management Module, OTP & Email Module
Security Features	Protected API routes, JWT token verification, OTP-based password reset, secure file upload handling

2 Introduction to Tools

The DP Tradeking HRMS & Payroll is developed using modern full-stack technologies under the MERN stack architecture.

MongoDB

MongoDB is a NoSQL database used to store employee, department, attendance, HR, and admin data. It stores data in JSON-like documents (BSON format). MongoDB provides:

- Flexible schema design
- High performance
- Scalability
- Easy integration with Node.js

Collections Used:

- users (Admin)
- hrlogin
- employees
- departments
- attendance

Express.js

Express.js is a backend framework used to build RESTful APIs. It handles:

- HTTP requests and responses
- Middleware implementation
- Route handling
- Authentication logic

It connects frontend requests to database operations.

React.js

React.js is used to develop the frontend user interface. It provides:

- Component-based architecture
- Reusable UI components
- State management using hooks
- Responsive and dynamic UI

DP Tradeking HRMS & Payroll System

Pages Developed:

- Login Page
- Dashboard (Admin / HR / Employee)
- Employee Management
- Department Management
- Attendance Management

Node.js

Node.js is the runtime environment used to execute backend JavaScript code. It allows:

- Asynchronous programming
- Fast server-side processing
- Integration with MongoDB

JWT (JSON Web Token)

JWT is used for secure authentication. After login:

- A token is generated
- The token is stored in frontend
- Every request is verified using middleware

This ensures secure role-based access.

Multer

Multer is used for file uploads:

- Profile photos
- Resume documents

Files are stored in the uploads folder.

Node mailer

Node mailer is used for:

- Sending OTP for password reset
- Sending login credentials to employees

3 System Study

3.1 Existing System

The existing system in many small and medium organizations is manual or spreadsheet-based. The limitations include:

1. Data Redundancy
Duplicate entries may occur.
2. Lack of Security
No proper access control mechanism.
3. Time-Consuming
Manual attendance tracking consumes time.
4. No Real-Time Updates
Data updates are not reflected immediately.
5. No Backup Management
Risk of data loss.
6. Limited Reporting
Hard to generate structured reports.

3.2 Proposed System

The proposed system is a web-based HR Management application that solves existing problems.

Features of Proposed System:

- Centralized database
- Role-based authentication (Admin, HR, Employee)
- Employee CRUD operations
- Department management
- Attendance tracking
- OTP-based password reset
- Email notifications
- Secure JWT authorization
- File upload support

Benefits:

- Reduced paperwork
- Increased efficiency
- Improved data security
- Real-time updates
- Structured data management

3.3 Scope, Aim, and Objectives

Scope

The scope of the project includes:

- Managing employee records digitally
- Managing departments
- Tracking attendance
- Role-based login system
- Generating reports (basic)
- Secure authentication

The system is scalable and can be extended in the future to include:

- Payroll Management
- Leave Management
- Performance Evaluation
- Report generation module

3.4 Aim & Objective

The main aim of this project is to design and develop a secure, scalable, and efficient Employee Management System using MERN stack technology.

Objectives

1. To implement secure login with JWT authentication.
2. To create a centralized employee database.
3. To automate attendance management.
4. To manage department structure.
5. To provide OTP-based password reset.
6. To ensure role-based access control.
7. To reduce manual HR workload.

3.5 Feasibility Study

3.5.1 Technical Feasibility

The project uses modern and widely adopted technologies:

- MERN stack is scalable and reliable.
- MongoDB provides flexible schema management.
- Node.js ensures fast backend processing.
- JWT ensures secure authentication.

The system can run on standard hardware and does not require expensive infrastructure.

Conclusion: Technically feasible.

3.5.2 Economic Feasibility

- All tools used are open-source.
- No license cost required.
- Development cost is low.
- Maintenance cost is minimal.

Conclusion: Economically feasible.

3.5.3 Operational Feasibility

- User-friendly interface.
- Simple dashboard navigation.
- Minimal training required.
- Improves HR efficiency.

Conclusion: Operationally feasible.

4. Requirement Analysis

4.1 Functional Requirements

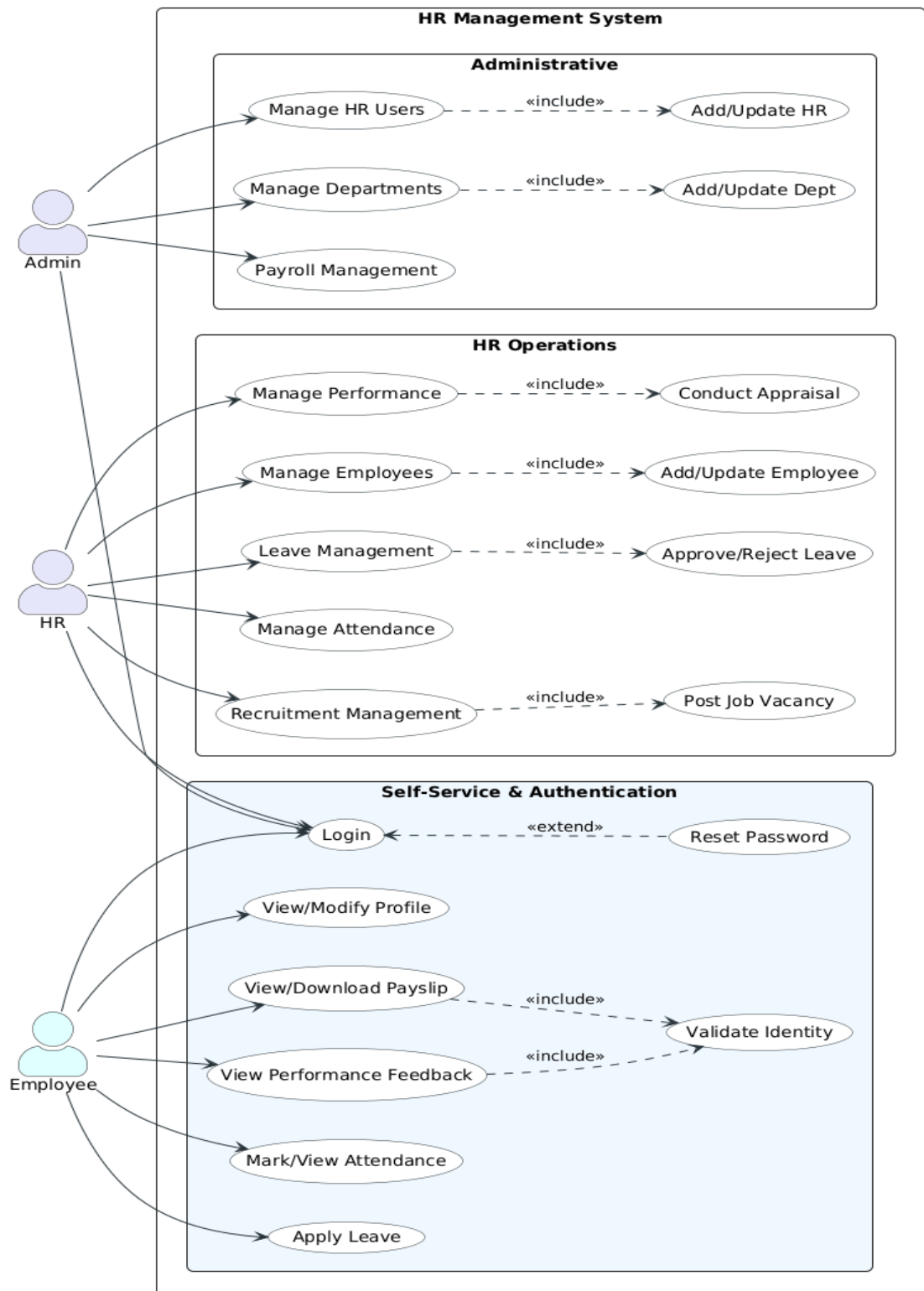
1. User Authentication
 - a. Admin login

- b. HR login
 - c. Employee login
- 2. Employee Management
 - a. Add employee
 - b. Update employee
 - c. Delete employee
 - d. View employee list
- 3. Department Management
 - a. Add department
 - b. View department
 - c. Activate/Deactivate department
- 4. Attendance Management
 - a. Mark attendance
 - b. View attendance
 - c. Delete attendance
- 5. Password Management
 - a. Forgot password
 - b. OTP verification
 - c. Reset password
- 6. Email System
 - a. Send credentials
 - b. Send OTP

4.2 Non-Functional Requirements

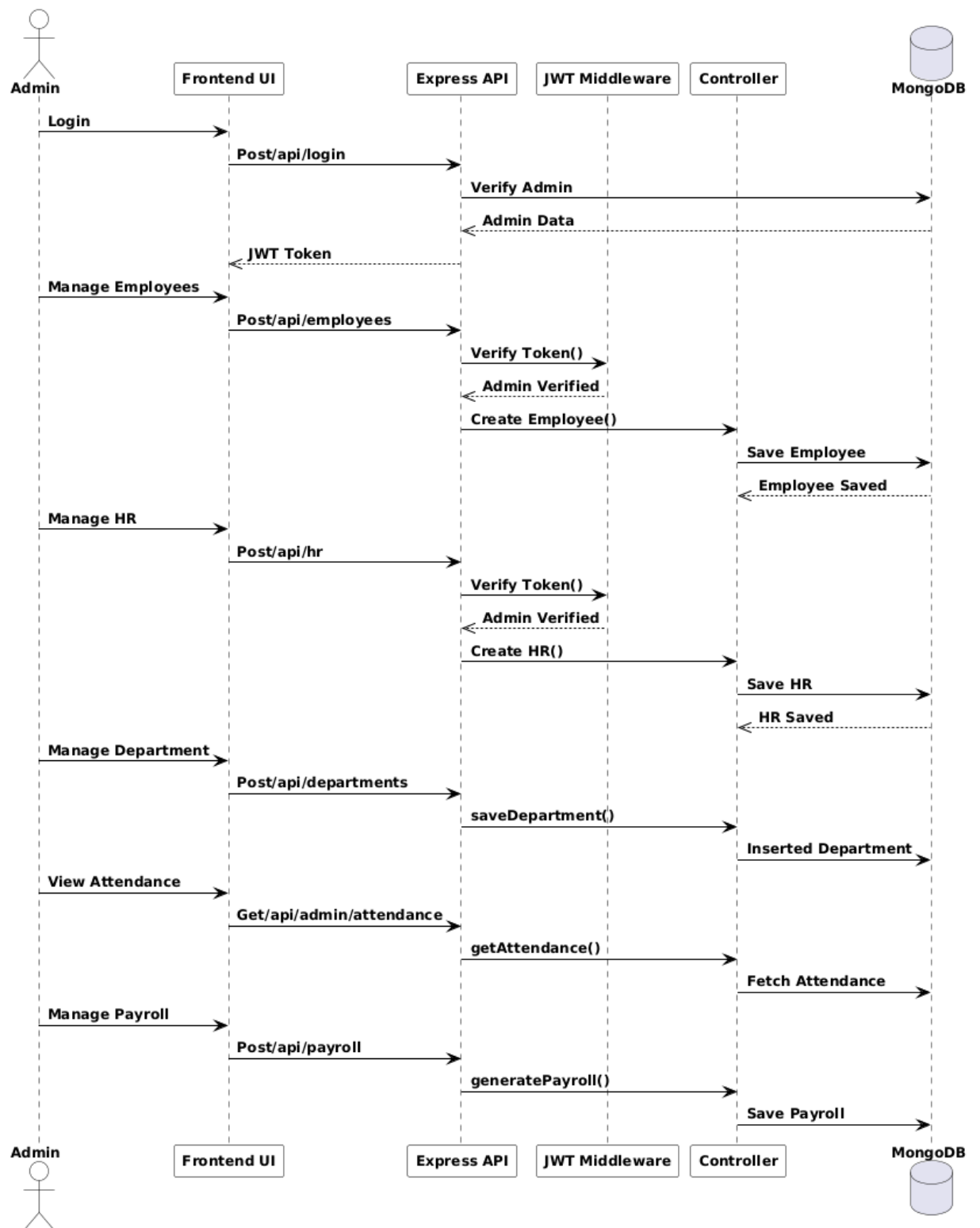
- 1. Security
 - a. JWT-based authentication
 - b. Role-based authorization
- 2. Performance
 - a. Fast response time
 - b. Efficient database queries
- 3. Scalability
 - a. System can handle increased data
- 4. Usability
 - a. Easy navigation
 - b. Responsive UI
- 5. Reliability
 - a. Proper error handling
 - b. Data validation

4.3 Use Case Diagram

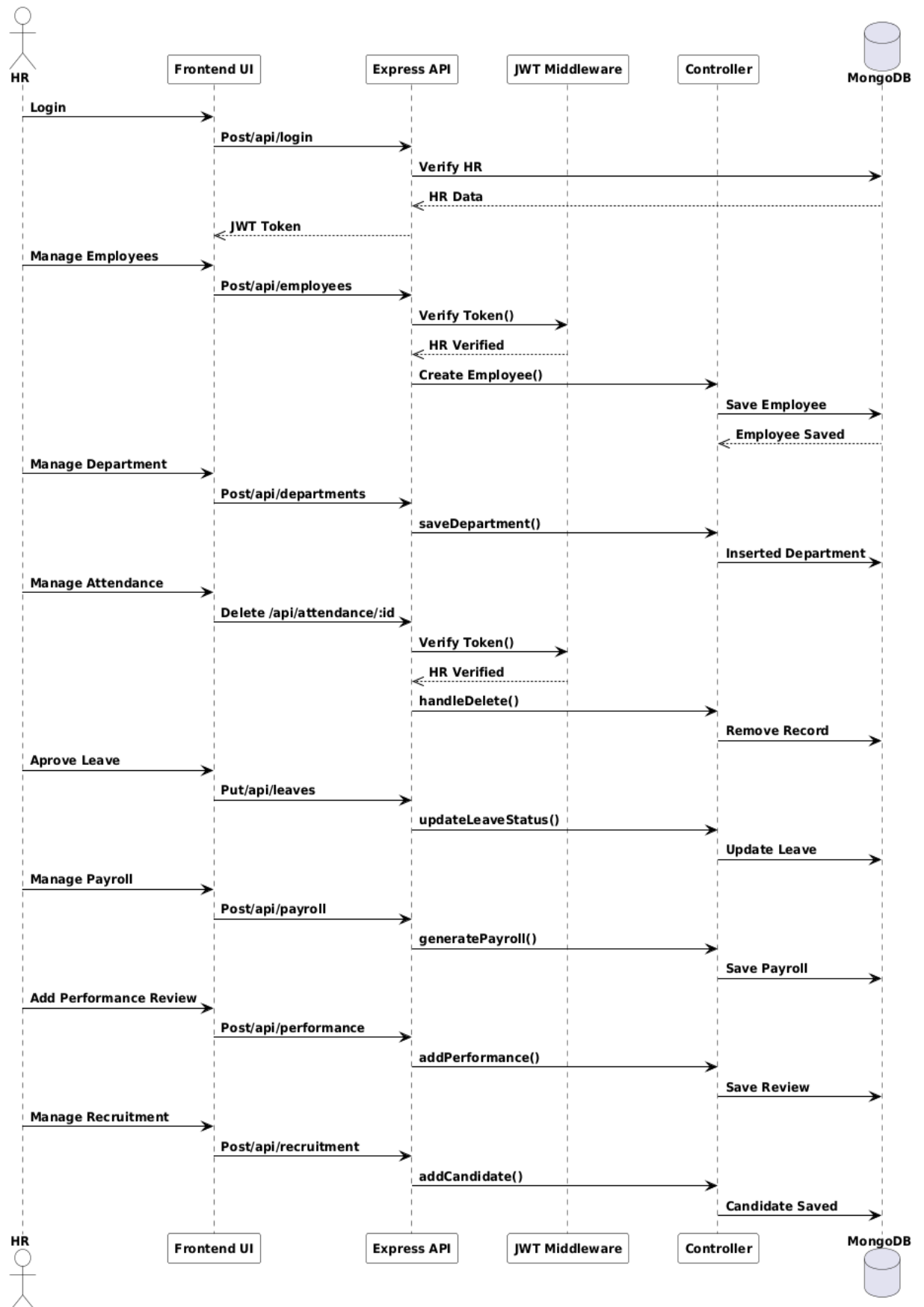


4.4 Sequence Diagram

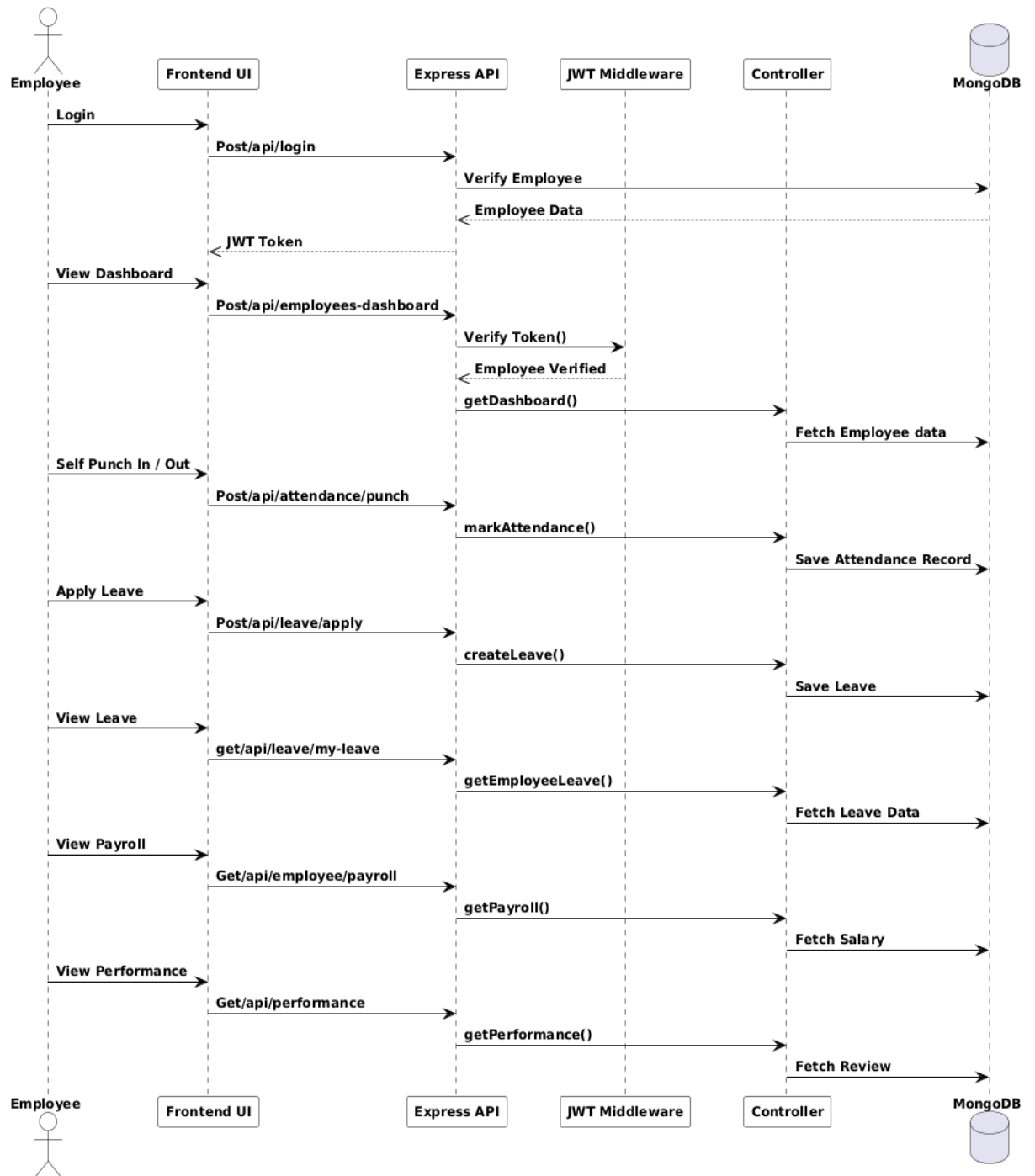
4.4.1 Admin Sequence



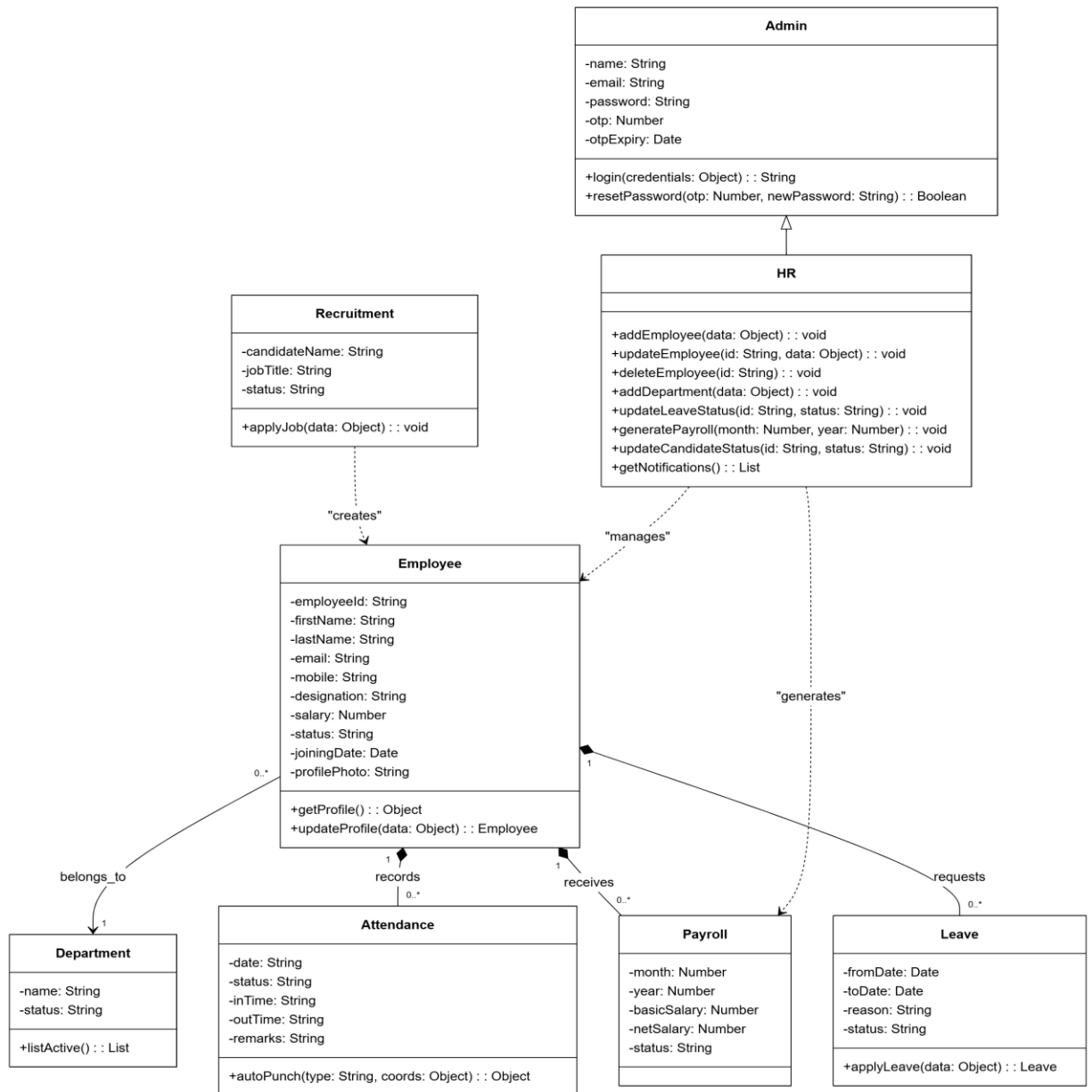
4.4.2 HR Sequence



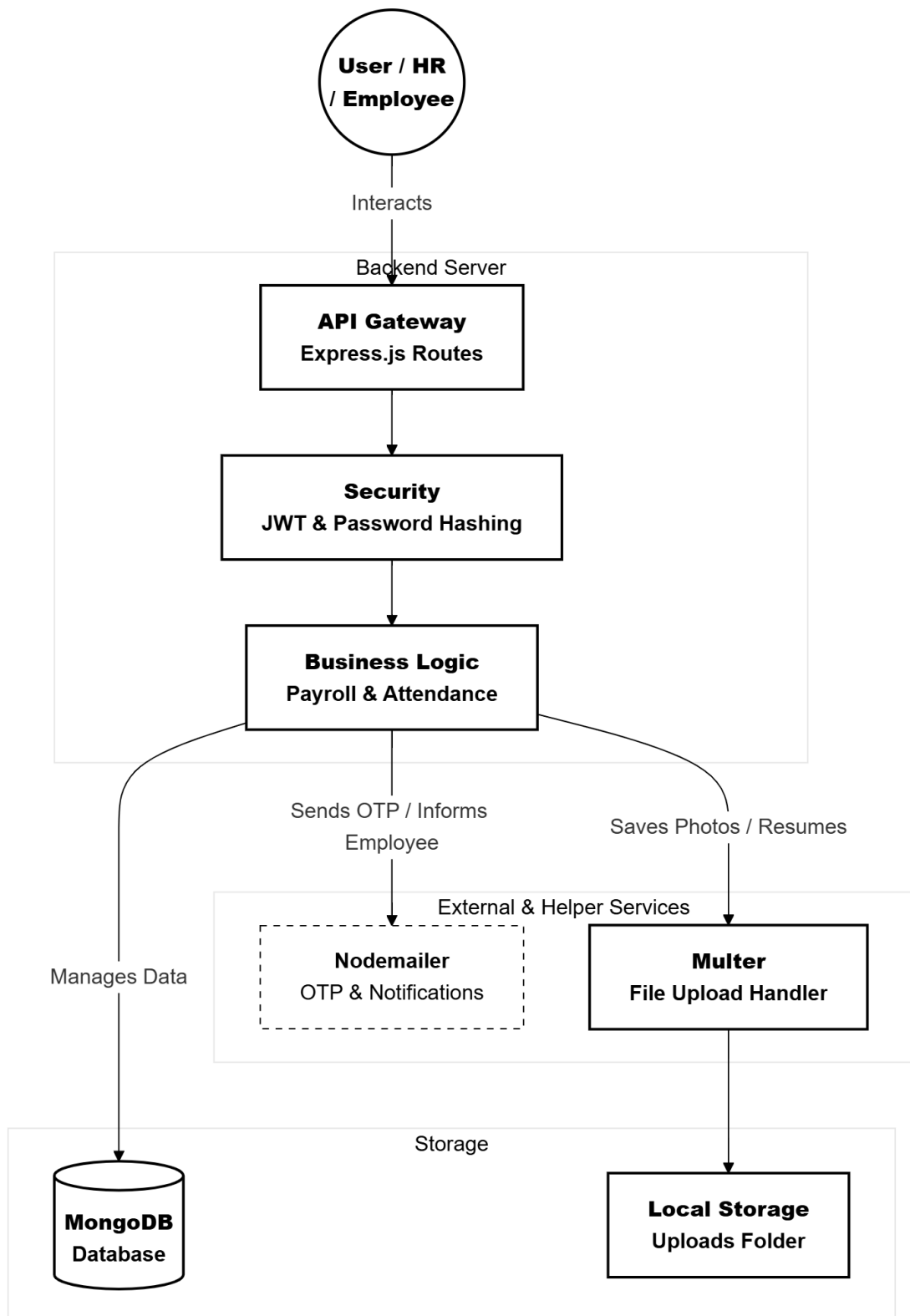
4.4.3 Employee Sequence



4.5 Class Diagram



4.6 System Architecture Diagram



5 System Design

5.1 Data Dictionary

1 Users Table:

Field Name	Data Type	Description	Notes
_id	ObjectId	Default unique identifier generated by MongoDB	Automatically created
name	String	Full name of the user	Required
email	String	User email address	Required
password	String	User password	Required
otp	Number	One-time password	Optional
otpExpiry	Date	OTP expiration timestamp	Optional
createdAt	Date	Record creation time	Auto-generated
updatedAt	Date	Record update time	Auto-generated

2 HR Table:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
name	String	HR name	Required
email	String	HR email	Required
password	String	HR password	Required

3 Departments:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
name	String	Department name	Unique Index
status	String	Department status	Default: Active
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

4 Employees:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
employeeId	String	System-generated employee code	Auto-generated DPTK001
firstName	String	First name	Required
lastName	String	Last name	Optional
email	String	Employee email	Unique Index
mobile	String	Contact number	Optional
gender	String	Gender	Optional
department	ObjectId	Reference for Department collection	Stored as ObjectId
designation	String	Job title	Optional
joiningDate	Date	Date of joining	Optional
salary	Number	Basic salary	Required
status	String	Employment status	Default: Active
password	String	Login password	Required
address	String	Residential address	Optional
dob	Date	Date of birth	Optional
documents	Object	Nasted Document Object	Require for new employee join
profilePhoto	String	Profile image path	Optional
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

5 Leave Table:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
employeeId	ObjectId	Reference to employees collection	Indexed
fromDate	Date	Leave start date	Required
toDate	Date	Leave end date	Required
reason	String	Leave reason	Required
status	String	Pending / Approved / Rejected	Default: Pending
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

6 Attendances:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
employeeId	ObjectId	Reference to employees collection	Indexed
date	Date	Attendance date	Required
status	String	Present / Absent / Half-Day / Leave	Enum
inTime	String	Check-in time	Optional
outTime	String	Check-out time	Optional
lat / lng	Number	GPS Coordinates	Captured at punch-in to verify if the user is within 2km of the office.
remarks	String	Additional notes	Optional
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

7 Payroll Settings:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
hraPercent	Number	HRA percentage	Default: 20
daPercent	Number	DA percentage	Default: 10
pfPercent	Number	PF percentage	Default: 12
taxPercent	Number	Tax percentage	Default: 5
latePenaltyGrace	Number	Late mark threshold	Default: 3. For every 3 late marks, 0.5 days of salary is deducted.
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

8 Payroll:

Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
employeeId	ObjectId	Reference to employees collection	Indexed
month	Number	Payroll month	Required
year	Number	Payroll year	Required
basicSalary	Number	Base salary	Calculated
hra	Number	HRA amount	Calculated
da	Number	DA amount	Calculated
grossSalary	Number	Gross salary	Calculated
pf	Number	PF deduction	Calculated
tax	Number	Tax deduction	Calculated
attendanceDeduction	Number	Deduction due to absence	Calculated
otherDeductions	Number	Additional deductions	Default: 0
bonus	Number	Bonus amount	Default: 0
presentDays	Number	Present days count	Calculated
absentDays	Number	Absent days count	Calculated
halfDays	Number	Half days count	Calculated
totalDeductions	Number	Total deductions	Calculated
netSalary	Number	Final salary	Calculated
status	String	Generated / Paid	Default: Generated
paymentDate	Date	Payment date	Optional
createdAt	Date	Creation time	Auto-generated
updatedAt	Date	Update time	Auto-generated

9 Recruitment:

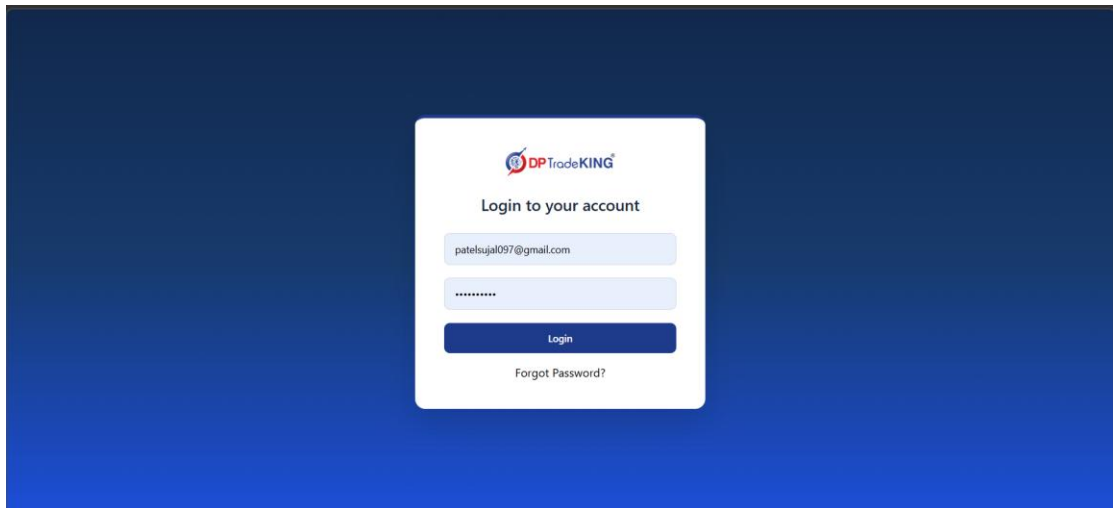
Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
name	String	Candidate name	Required
email	String	Candidate email	Required
position	String	Applied position	Required
status	String	Applied / Shortlisted / Selected / Rejected	Default: Applied
interviewDate	Date	Interview date	Optional
remarks	String	HR remarks	Optional

10 Performance:

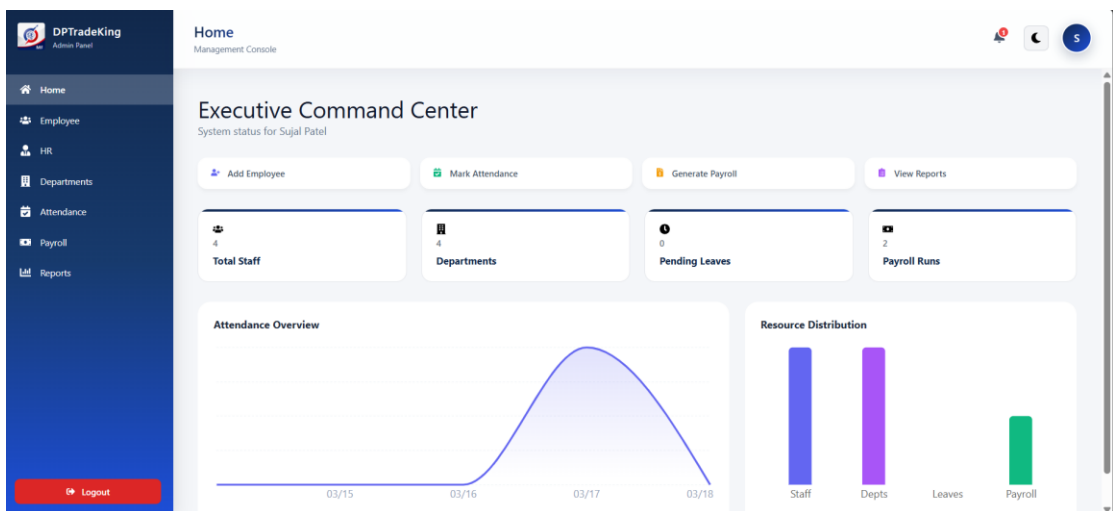
Field Name	Data Type	Description	Notes
_id	ObjectId	Unique identifier	Auto-generated
employeeId	ObjectId	Reference to employees collection	Indexed
reviewPeriod	String	Review period	Required
workQuality	Number	Rating (1–5)	Required
productivity	Number	Rating (1–5)	Required
behavior	Number	Rating (1–5)	Required
initiative	Number	Rating (1–5)	Required
kpiTarget	Number	KPI target value	Required
kpiAchieved	Number	KPI achieved value	Required
finalRating	Number	Computed performance rating	Calculated
kpiScore	Number	KPI percentage score	Calculated
attendanceScore	Number	Attendance-based score	Calculated
incrementSuggestion	Number	Suggested increment (%)	Optional
comments	String	HR remarks	Optional

5.2 Screen Layouts /UI Design

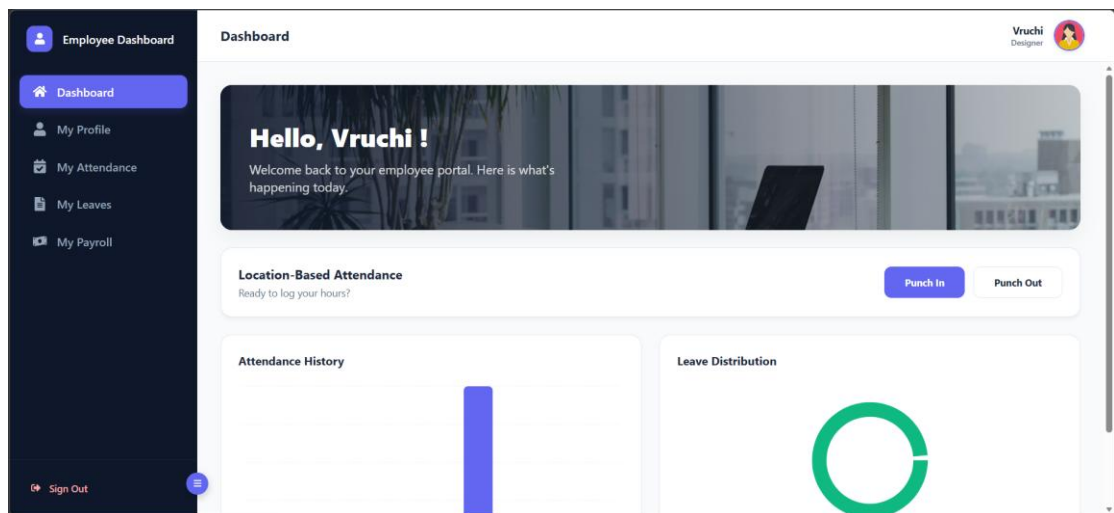
1 Login Page:



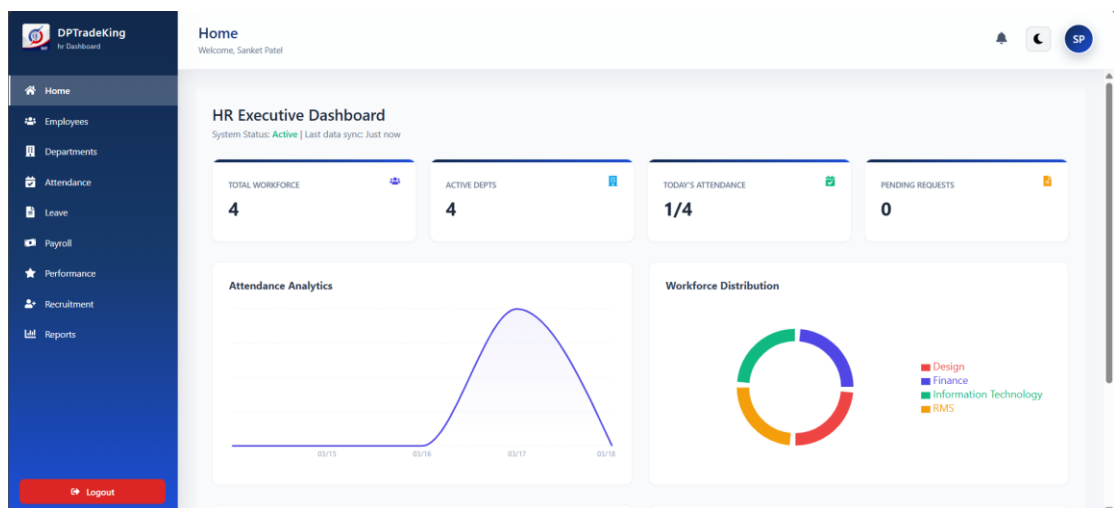
2 Admin Dashboard:



3 Employee Dashboard



4 HR Dashboard



5 Employee Module:

DPTradeKing
for Panel

Employees
Welcome back, Sanket Patel

All Employees

+ Add Employee

Search... All

ID	NAME	EMAIL	DESIGNATION	DEPARTMENT	STATUS	ACTIONS
EMP001	Sunil Patel	patelsunil098@gmail.com	Employee	Mutual Fund	Active	View Edit
EMP002	Amit Patel	patelamit@gmail.com	IT	Finance	Active	View Edit
EMP003	Sumit Kumar	patelsunit@gmail.com	ERR	Finance	Active	View Edit
EMP004	Jay Thakor	Thakorjay@gmail.com	Employee	Information Technology	Active	View Edit
EMP005	Punitkumar Patel	punitpatel2764@gmail.com	CCO	Mutual Fund	Active	View Edit
EMP006	Kishan Patel	patelkishan@gmail.com	Networking	Information Technology	Active	View Edit

Logout

6 Add Employee:

Add Employee

First Name:

Last Name:

Email:

Mobile Number:

Gender:

Department:

Designation:

Joining Date:

Date of Birth:

Salary:

Password:

Address:

Role:

Status:

7 Department:

Departments

Department Name:

Active:

Add Department:

Search Department:

NAME	STATUS	CREATED AT	ACTION
Mutual Fund	Active	19/2/2026	Delete
Finance	Active	20/2/2026	Delete
Information Technology	Active	20/2/2026	Delete
RMS	Active	20/2/2026	Delete

8 Attendance:

Attendance

CURRENTLY ONLINE: 2 Employees

Master Attendance Log

Search employee:

dd-mm-yyyy

Reset Filters:

Export Data:

EMPLOYEE & STATUS	DATE	STATUS	PUNCH IN	PUNCH OUT	FLAGS & REMARKS	ACTION
Kishan Patel OFFLINE	2026-03-16	Half-Day	14:39	15:31	In: Office Out: Location recorded	Delete
Sumit Kumar LOGGED IN	2026-03-16	Half-Day	14:36	Active	In: Office	Delete
Vrush Parmar LOGGED IN	2026-03-16	Half-Day	14:36	Active	In: Office	Delete

9 Leave:

Leave Management

Search by employee...

EMPLOYEE	FROM	TO	REASON	STATUS	APPLIED ON	ACTIONS
Punitkumar Patel	28/2/2026	3/3/2026	Holi	Rejected	25/2/2026	Delete
Sumit Kumar	26/2/2026	2/3/2026	Family Function	Approved	25/2/2026	Delete
Punitkumar Patel	25/2/2026	27/2/2026	Function	Rejected	25/2/2026	Delete
Sumit Kumar	19/2/2026	25/2/2026	Sick	Approved	22/2/2026	Delete

10 Payroll:

Payroll Management

Request Settings Update

Select Employee Month Year Bonus Other Deductions Generate Payroll

EMPLOYEE	MONTH	YEAR	GROSS	NET	STATUS	ACTIONS
Vrush Parmar	March	2026	₹ 3000	₹ 2573.08	Paid	Payroll

11 Performance:

Performance Management

Auto-Generate Performance

EMPLOYEE	REVIEW PERIOD	ATTENDANCE	WORK QUALITY	BEHAVIOR	INITIATIVE	FINAL RATING	GRADE	INCREMENT %	ACTION
Kishan Patel (Network Manager)	3	2	3	3	3	2.7	C	0%	Delete
Sumit Kumar (Analysier)	3	2	3	3	3	2.7	C	0%	Delete
Vrush Parmar (Designer)	3	2	3	3	3	2.7	C	0%	Delete

12 Recruitment:

Recruitment Management

Welcome back, Sanket Patel

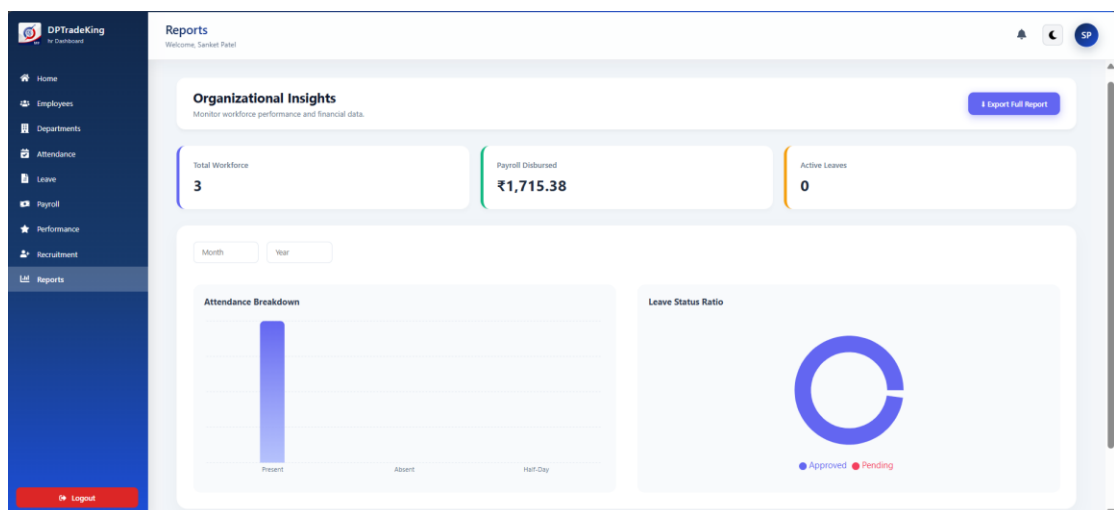
Candidate Name: Candidate Email: Position Applied For: Applied: dd-mm-yyyy:

Remarks: [Add Candidate](#)

NAME	EMAIL	POSITION	STATUS	INTERVIEW DATE	REMARKS	ACTION
Punit	punitpatel2764@gmail.com	Employee	Interview Scheduled	2026-03-08		Edit Delete
vruchi	vruchiparmar912@gmail.com	Employee	Selected	2026-03-06		Edit Delete

[Logout](#)

13 Reports:



14 Employee Profile:

Employee Dashboard

My Profile

Vruchi Designer

My Profile [Edit Profile](#)

First Name: Last Name: Email:

Department:

Department change requires HR approval

[Sign Out](#)

15 Employee Attendance:

My Attendance

[View Attendance Report](#)

DATE	STATUS	IN TIME	OUT TIME
16/3/2026	Half Day	14:36	-

16 Employee Leave Apply:

Apply for Leave

From Date To Date

Reason

[Apply Leave](#)

My Leave History

FROM	TO	REASON	STATUS
13/3/2026	16/3/2026	Going to Tracking	Approved

17 Employee Salary Slip:

My Payroll

MONTH	YEAR	NET SALARY	STATUS	PAYSリップ
3	2026	₹ 2,573.08	Paid	Download

6 System Implementation

The system is implemented using the **MERN (MongoDB, Express, React, Node.js)** stack. The backend is built with Node.js and Express, utilizing Mongoose for data modeling. Security is handled via JWT (JSON Web Tokens) and Bcrypt for password hashing. External communications (OTPs and Notifications) are integrated using Nodemailer.

6.1 Module Implementation

The system is divided into five core functional modules:

1. Authentication & Security Module

- This module acts as the gatekeeper. It handles user identity and protected access.
- **Key Logic:** Uses jsonwebtoken (JWT) to create secure sessions.
- **Implementation: * Login:** Validates credentials and returns a token.
 - **Password Reset:** Generates a 6-digit **OTP (One-Time Password)** and sends it via email to verify the user before allowing a password change.

2. Employee Management Module

This is the central data hub where employee profiles are created and maintained.

- **Key Logic:** Handles CRUD (Create, Read, Update, Delete) operations for employees and departments.
- **Implementation:** Uses **Multer** middleware to process and store physical files such as **Profile Photos** and **Resume PDFs** in the local file system while storing the file paths in MongoDB.

3. Attendance & Tracking Module

This module tracks daily presence and location-based validation.

- **Key Logic:** Implements "Punch-In" and "Punch-Out" logic.

- **Implementation:** It captures the date, time, and **GPS coordinates (Latitude/Longitude)**. It includes a distance check to ensure employees are within the allowed radius of the office location.

4. Payroll & Salary Module

This module automates the financial calculations for the organization.

- **Key Logic:** Calculates Net Salary based on Basic Pay, HRA, and Deductions.
- **Implementation:** It filters attendance records for a specific month to determine "Paid Days." Once the HR generates the payroll, the status is set to "Generated" or "Paid."

5. Notification & Communication Module

This module keeps the staff informed and the workflow moving.

- **Key Logic:** Uses **Nodemailer** for automated messaging.
- **Implementation: * OTP Service:** Sends security codes for forgotten passwords.
 - **Employee Information:** Sends email alerts to **inform employees** when their leave requests are approved/rejected or when their monthly payroll is generated.
 - **Dashboard Alerts:** Aggregates pending tasks (like pending leaves or new job applications) for the HR manager's view.

6. Recruitment Module

Handles the intake of new talent into the organization.

- **Key Logic:** Manages candidate applications and hiring status.
- **Implementation:** Stores candidate details and allows HR to update their status from "Applied" to "Interviewing" or "Hired." Once hired, the data can be transitioned into the Employee module.

7. System Testing

System testing ensures that the complete HR Management System functions as expected according to the requirements. It verifies the integration of all modules, the correctness of data flow, security measures, and user role-based functionality.

7.1 Testing Strategies

The following testing strategies were applied:

1. Functional Testing

- Each module (Employee, HR, Department, Attendance, Leave, Payroll) was tested for CRUD operations.
- Role-based access control was verified for Admin, HR, and Employee users.
- Validation of input fields (e.g., mandatory fields, unique email, password length) was tested.

2. Integration Testing

- Verified integration between the backend APIs and frontend components.
- Tested the interaction of Attendance and Leave modules with Payroll calculation.
- Ensured that employee data, attendance records, and payroll records are correctly linked.

3. User Interface Testing

- Checked responsive design of dashboards and lists.
- Verified visibility of buttons and status badges for different roles.
- Ensured proper display of employee profiles, payslips, and attendance summary.

4. Security Testing

- Validated JWT authentication and role-based access to APIs.
- Tested password reset functionality with OTP verification.

- Ensured sensitive data like passwords are not exposed in API responses.

5. Boundary and Negative Testing

- Attempted invalid inputs such as duplicate emails, missing mandatory fields, or invalid dates.
- Verified the system prevents duplicate attendance or payroll generation for the same period.
- Checked for handling of unauthorized access attempts.

6. Performance Testing

- Simulated multiple concurrent users (HR, Admin, Employees) performing CRUD operations.
- Verified the system maintains acceptable response times under load.

7.2 Test Cases:

ID	Module	Test Scenario	Test Steps	Expected Result
TC-01	Authentication	Admin Login	Enter valid admin email & password → Click Login	Admin logged in → Dashboard displayed
TC-02	Authentication	Invalid Login	Enter wrong credentials → Click Login	Error: "Invalid Email or Password"
TC-03	Employee Management	Add Employee	Fill all mandatory fields → Submit	Employee created with unique Employee ID
TC-04	Employee Management	Add Employee with Existing Email	Enter duplicate email → Submit	Error: "Email already exists"
TC-05	Employee Management	Update Employee	Edit employee details → Submit	Employee details updated successfully
TC-06	Employee Management	Delete Employee	Delete employee record	Employee deleted successfully (Admin)

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ID	Module	Test Scenario	Test Steps	Expected Result
TC-07	Attendance	Mark Attendance	Select employee, date, status → Submit	Attendance added successfully
TC-08	Attendance	Duplicate Attendance	Submit attendance for same employee & date	Error: "Attendance already marked for this date"
TC-09	Leave	Apply Leave	Employee submits leave request	Leave saved with "Pending" status
TC-10	Leave	Approve Leave	Admin approves leave	Leave status changes to "Approved"
TC-11	Payroll	Generate Payroll	Admin selects employee, month, year → Submit	Payroll generated based on attendance & leaves
TC-12	Payroll	Delete Paid Payroll	Try to delete payroll marked "Paid"	Error: "Cannot delete Paid payroll"
TC-13	Profile	View Profile	Employee accesses own profile	Profile displayed without password
TC-14	File Upload	Upload Resume / Photo	Add employee with files	Files saved in /uploads and path stored in DB

8 Future Enhancement

While the current HR Management System provides a comprehensive solution for employee, attendance, leave, and payroll management, several improvements can be implemented to enhance its functionality, scalability, and user experience:

1. Advanced Security Features

- **Password Hashing & Encryption:** Currently, passwords are stored in plain text. Implementing hashing algorithms like bcrypt will improve security.
- **Two-Factor Authentication (2FA):** Adding OTP-based login or Google Authenticator integration for sensitive roles like Admin and HR.
- **Role-Based Access Control (RBAC):** Fine-grained permissions to restrict access to specific modules, actions, and sensitive data.

2. Enhanced Payroll & Compensation Management

- **Automated Tax Calculation:** Integration with government tax APIs for accurate tax computation.
- **Salary Components Customization:** Ability to define allowances, deductions, and bonuses per employee dynamically.
- **Salary History & Comparison:** Visual charts to compare salaries across months or departments.

3. Employee Self-Service Portal

- **Profile Management:** Employees can update personal information, upload documents, and view performance metrics.
- **Leave & Attendance Requests:** Real-time tracking and approval workflows with notifications.
- **Payslip Access:** Secure online generation and download of payslips, with historical records.

4. Advanced Attendance Management

DP Tradeking HRMS & Payroll System

- **Biometric Integration:** Support for fingerprint or facial recognition to log attendance automatically.
- **Shift & Overtime Tracking:** Capture shifts, overtime, and flexible work hours.
- **Late & Early Notifications:** Automatic alerts for late arrivals or early departures.

5. Analytics & Reporting

- **Dashboard Insights:** Enhanced visualizations for HR analytics, including employee retention, absenteeism trends, and department-wise performance.
- **Export Reports:** PDF, Excel, and CSV exports for payroll, attendance, and leaves.
- **Predictive Analytics:** Use machine learning to predict employee attrition or performance trends.

6. Integration & Automation

- **Email & SMS Notifications:** Automated reminders for pending leaves, payroll release, and attendance irregularities.
- **Third-Party Integration:** Connect with tools like Slack, Google Workspace, or accounting software for seamless workflows.
- **API for Mobile Apps:** Enable mobile access for employees and managers via REST or GraphQL APIs.

7. Mobile & Cross-Platform Support

- **Mobile Application:** Dedicated app for employees and HR for easier access on the go.
- **Push Notifications:** Instant updates on leave approvals, payroll status, and announcements.

8. Document & Compliance Management

- Digital Document Storage: Central repository for contracts, resumes, appraisal letters, and certificates.
- Audit Trails: Track modifications for compliance and accountability.
- Policy Management: Upload and notify employees about HR policies.

9. Performance & Training Management

- Employee Appraisal System: Track KPIs, goals, and appraisals.
- Training & Skill Management: Manage training programs, certifications, and skill development tracking.

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